

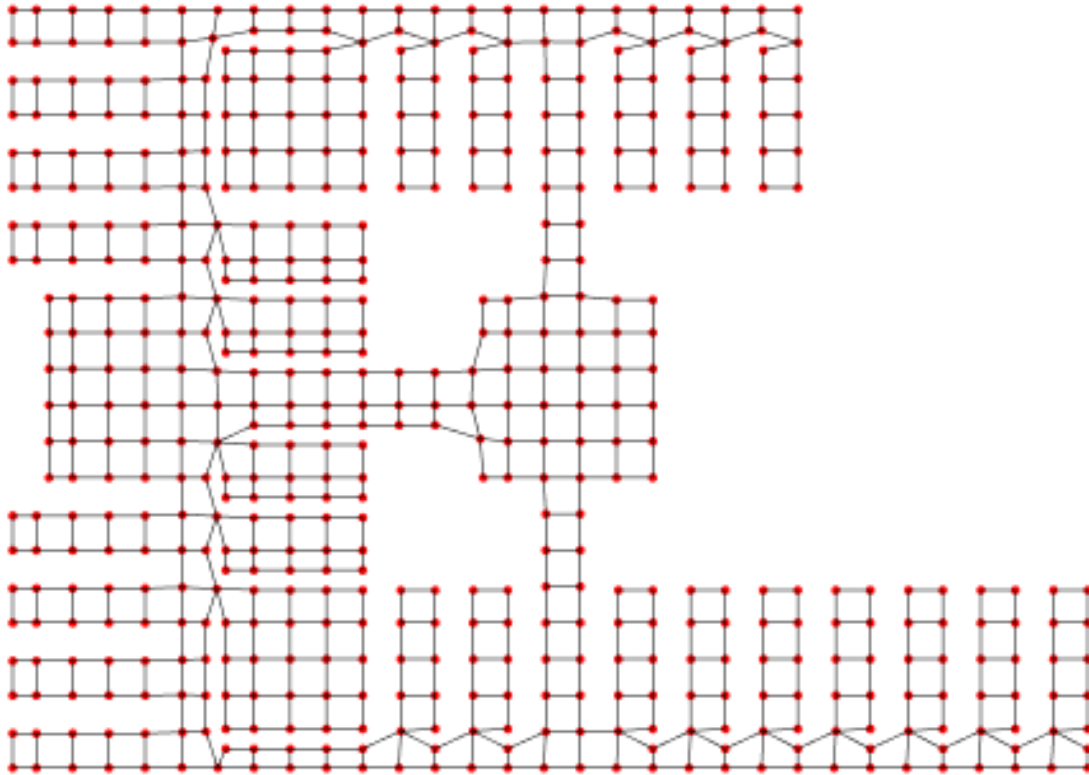
Graph Analysis of Architectural Structure

Assignment 2

Hotel Layout

Ahmad Al Baltaji

MINIMUM SPANNING TREE



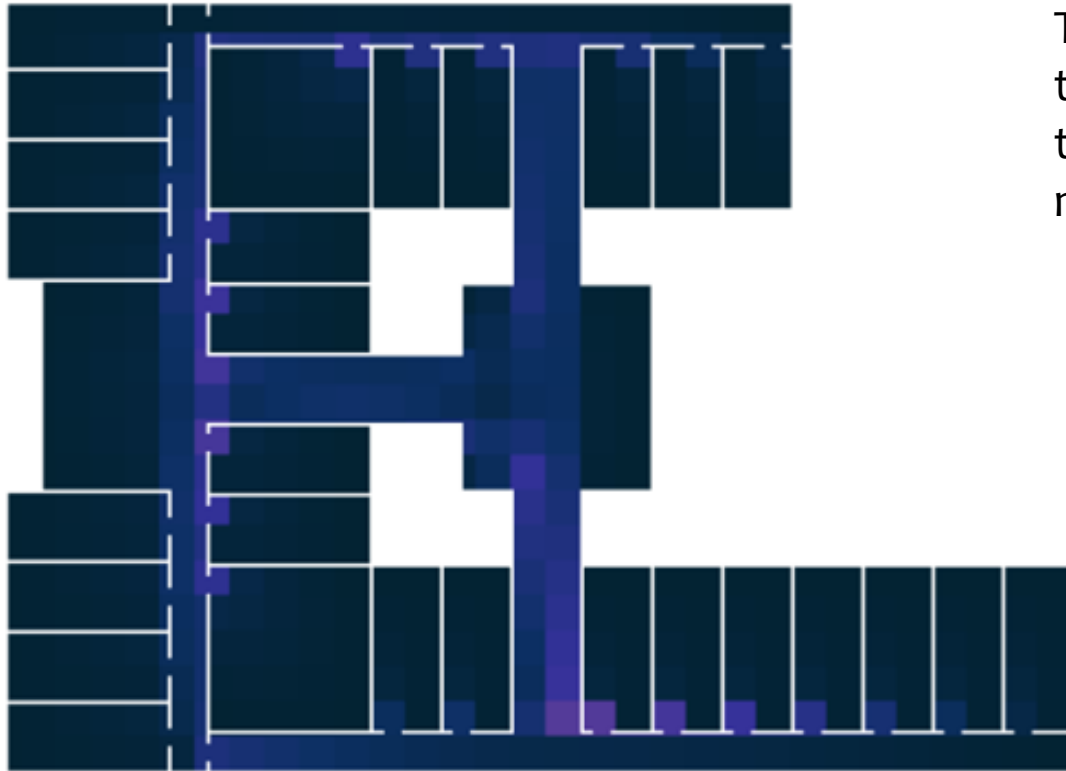
Ultimately, the MST proves the designed layout is an efficient, tree-like circulation system with modularity

SHORTEST PATH



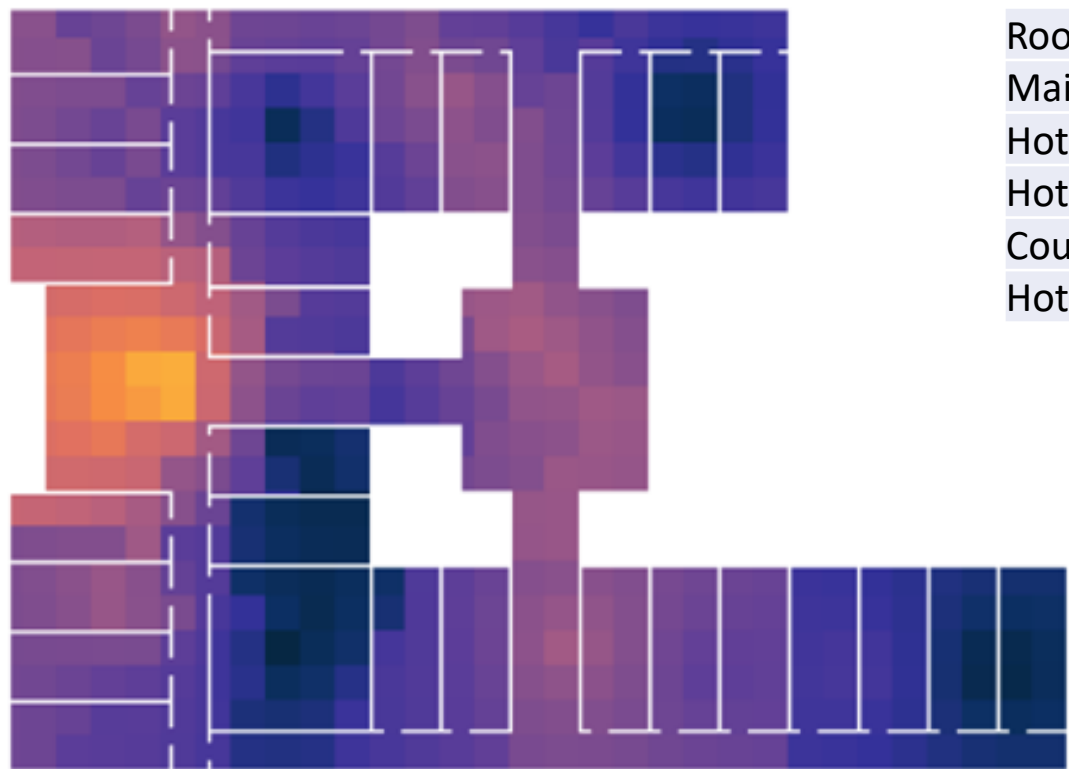
Shortest Path Duration: 0.47 seconds Straighten Wire
Duration: 103.8 seconds Original Shortest Path Length: 98.94
Straightened Shortened Path Length: 88.99

BETWEEN CENTRALITY



The Graph shows that each room is connected once to the corridor which are connected to each other then main central hall and the courtyard in the middle – Giving the correct accessibility and security

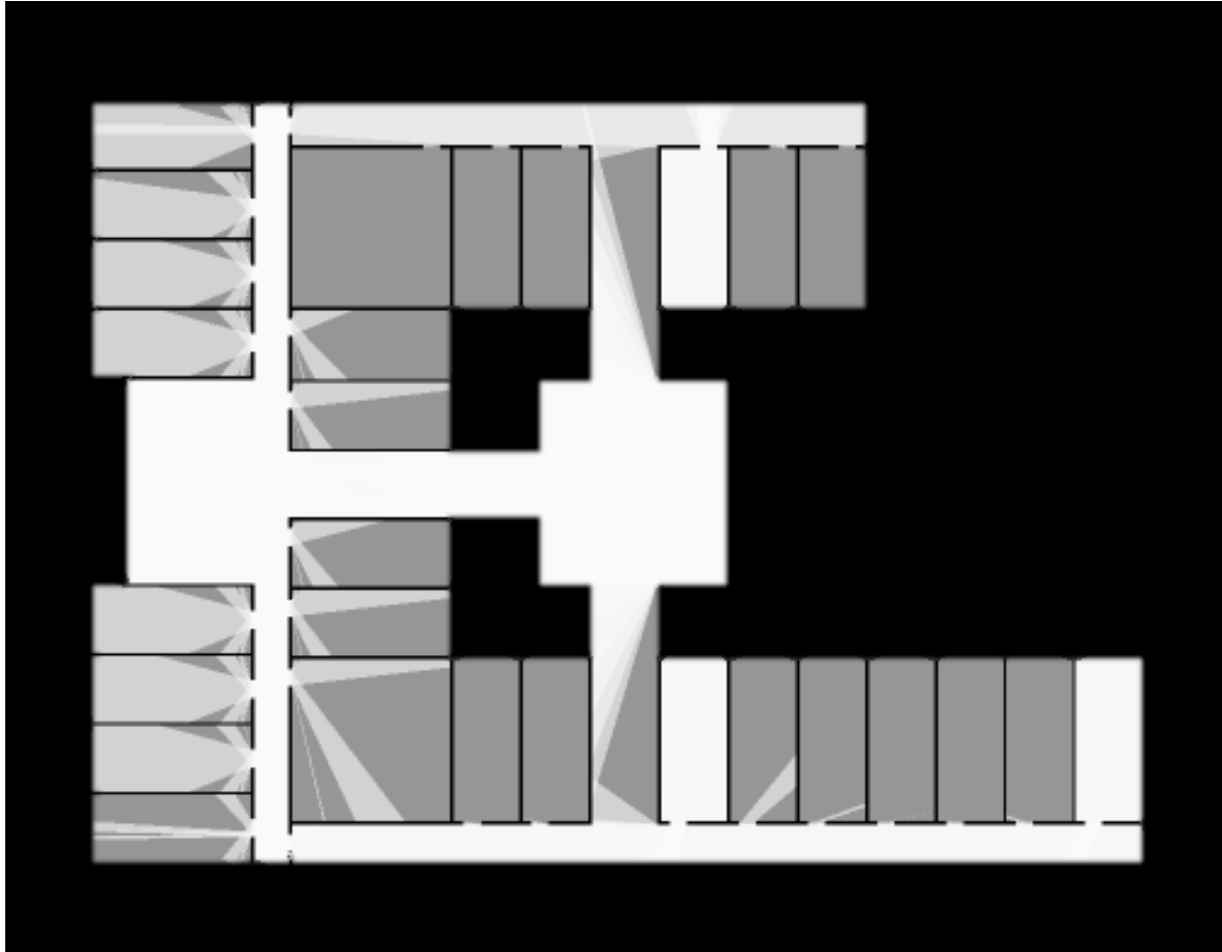
DEGREE CENTRALITY



Room Name	Degree Centrality	Num Connections
Main Hall	0.2778	5
Hotel Room	0.0556	1
Hotel Room	0.0556	1
Courtyard	0.1667	3
Hotel Room	0.0556	1

The graph shows most accessed spaces versus the least access / private spaces

Visibility Graphics



Visibility we can see the main hall and Courtyard are visible where as the corridors are less and least are the rooms giving the optimal privacy

Summary

According to each graph we can say:

1. The layout is modular easier for construction and implementation
2. The shortest path shows that it is abiding with the code
3. No bottlenecks shown and all rooms are connected for security and accessibility reasons
4. Main rooms which are the most connected are as expected the main entrance and the courtyard
5. Visibility we can see the main hall and Courtyard are visible where as the corridors are less and least are the rooms giving the optimal privacy